

CSE 400 – Fundamentals of Probability in Computing

Lecture 10: Randomized Min-Cut Algorithm

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1 Overview of the Lecture

This lecture studies the **minimum cut (min-cut) problem** in graphs and discusses both deterministic and randomized approaches for solving it. The lecture also presents the relationship between **maximum flow and minimum cut** in network flow problems.

1.1 Topics Covered

1. Min-Cut Problem
2. Applications of Min-Cut
3. Definition of Min-Cut
4. Successful Min-Cut Run
5. Unsuccessful Min-Cut Run
6. Max-Flow Min-Cut Theorem
7. Deterministic Min-Cut Algorithm
8. Stoer–Wagner Minimum Cut Algorithm
9. Randomized Min-Cut Algorithm
10. Karger’s Randomized Algorithm
11. Deterministic vs Randomized Comparison
12. Python Simulation Activity

2 The Min-Cut Problem

2.1 Motivation: Why Use Min-Cut?

The min-cut algorithm is used in several areas related to network analysis.

Step-by-step reasoning:

1. The min-cut algorithm helps analyze **network connectivity**.
2. It is useful in **reliability analysis** of networks.
3. It helps solve various **optimization problems**.

2.2 Applications

2.2.1 Network Design

- Helps improve communication efficiency.
- Optimizes network flow.
- Identifies the minimum capacity cut in a network.

2.2.2 Communication Networks

- Helps analyze network vulnerability to failures.
- Enables construction of robust and fault-tolerant communication systems.

2.2.3 VLSI Design

- Used for partitioning circuits into smaller components.
- Reduces interconnection complexity in Very Large Scale Integration (VLSI).

3 Definition of Min-Cut

3.1 Cut-Set in a Graph

Definition

A **cut-set** in a graph is defined as:

A set of edges whose removal breaks the graph into two or more connected components.

3.2 Minimum Cut Problem

Given a graph

$$G = (V, E)$$

where:

- V = set of vertices
- E = set of edges
- $n = |V|$ is the number of vertices

The **minimum cut problem** is defined as:

Finding a minimum cardinality cut-set in G .

3.3 Randomized Nature of Some Min-Cut Algorithms

Some min-cut algorithms (for example, **Karger's algorithm**) are randomized.

1. These algorithms depend on the order in which edges are selected.
2. If critical edges are contracted early, the algorithm may produce an incorrect cut.
3. Therefore, the algorithm has probabilistic behavior.

4 Edge Contraction Operation

Edge contraction is a fundamental operation used in many min-cut algorithms.

4.1 Definition

Edge contraction is defined as:

An operation that removes an edge from the graph while merging the two vertices connected by that edge.

4.2 Step-by-Step Description

When contracting an edge (u, v) :

1. Vertices u and v are merged into a single vertex.
2. Edges directly connecting u and v are removed.
3. All other edges remain unchanged.
4. The resulting graph may contain **parallel edges**.
5. The resulting graph does not contain **self-loops**.

This process gradually reduces the size of the graph while preserving possible cuts.

5 Successful Min-Cut Run

A **successful run** of a min-cut algorithm refers to the situation where:

1. The algorithm executes without contracting edges belonging to the true minimum cut.
2. The algorithm correctly identifies the minimum cut of the graph.

6 Unsuccessful Min-Cut Run

An **unsuccessful run** occurs when:

1. The algorithm contracts edges belonging to the true minimum cut too early.
2. As a result, the algorithm fails to identify the correct minimum cut.

7 Max-Flow Min-Cut Theorem

7.1 Theorem Statement

In a flow network, the maximum amount of flow passing from the source to the sink is equal to the total weight of the edges in a minimum cut.

7.2 Definitions

Capacity of a Cut

The sum of the capacities of edges oriented from a vertex in set X to a vertex in set Y .

Minimum Cut

The cut with the smallest possible capacity.

Maximum Flow

The largest possible flow from source S to sink T .

8 Deterministic Min-Cut Algorithm

8.1 Stoer–Wagner Minimum Cut Algorithm

Let s and t be vertices in graph G .

Let

$$G/\{s, t\}$$

denote the graph obtained by merging vertices s and t .

8.2 Theorem Used

A minimum cut of G can be obtained by taking the smaller of:

1. A minimum s - t cut of G
2. A minimum cut of $G/\{s, t\}$

8.3 Reasoning

Two cases may arise:

Case 1: A minimum cut separates s and t

- The minimum s - t cut of G is the answer.

Case 2: No minimum cut separates s and t

- The minimum cut of the contracted graph $G/\{s, t\}$ is the answer.

Algorithm 1 MinimumCutPhase(G, a)

```
 $A \leftarrow \{a\}$ 
while  $A \neq V$  do
    Add to  $A$  the most tightly connected vertex
end while
Return the cut weight (cut-of-the-phase)
```

Algorithm 2 MinimumCut(G)

```
while  $|V| \geq 1$  do
    Choose any vertex  $a \in V$ 
    Execute MinimumCutPhase( $G, a$ )
    if cut-of-the-phase is lighter than current minimum cut then
        Store it as the current minimum cut
    end if
    Shrink the graph by merging the two vertices added last
end while
Return the minimum cut
```

9 Deterministic Algorithm Pseudocode

10 Randomized Min-Cut Algorithm

10.1 Why Randomized Algorithms?

Randomized algorithms provide several advantages:

- Efficiency
- Parallelization
- Approximation guarantees
- Avoidance of worst-case instances
- Heuristic nature
- Robustness

11 Karger's Randomized Algorithm

The basic idea of the algorithm is as follows:

1. Randomly pick an edge.
2. Contract the selected edge.
3. Merge the two endpoints of the edge.
4. Repeat until only two vertices remain.
5. The remaining edges represent a cut of the graph.

The resulting cut may or may not be the minimum cut.

12 Comparison: Deterministic vs Randomized

12.1 Deterministic Min-Cut

- Always guarantees the exact minimum cut.
- May have higher time complexity for large graphs.

Example:

- Stoer–Wagner Algorithm

Time complexity:

$$O(V \cdot E + V^2 \log V)$$

12.2 Randomized Min-Cut

- Produces an approximate minimum cut with high probability.
- Uses random edge contractions.

Example:

- Karger's Algorithm

Time complexity:

$$O(V^2)$$

13 Theorem for Min-Cut Set

Theorem

The randomized algorithm outputs a min-cut set with probability at least

$$\frac{2}{n(n-1)}$$

where n is the number of vertices in the graph.

14 Python Simulation

Students are instructed to:

1. Open the Campuswire post for Lecture 10.
2. Download the provided `.ipynb` file.
3. Run the Python notebook to simulate the randomized min-cut algorithm.

15 Summary

This lecture introduced the **minimum cut problem** and its applications in network design, communication networks, and VLSI systems.

Two algorithmic approaches were discussed:

1. **Deterministic approach**

Stoer–Wagner algorithm, which guarantees the exact minimum cut.

2. **Randomized approach**

Karger’s algorithm, which uses random edge contractions to estimate the minimum cut with high probability.

The lecture also introduced the **Max-Flow Min-Cut Theorem** and concluded with a Python simulation activity.